

Multi-Stakeholder Collaborative Planning for Distribution Networks Based on Reinforcement Learning and Bi-level Game

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Abstract—With the rapid development of renewable energy and energy storage (ES) technologies, diversified market entities have emerged in distribution networks, forming a complex competitive landscape for electricity consumption. To address this challenge, this paper proposes a collaborative distribution network planning method that integrates reinforcement learning (RL) and bi-level game theory. Different from existing studies that only consider partial participants or adopt traditional heuristic algorithms, this paper establishes a multi-stakeholder game framework covering the distribution system operator (DSO), distributed generation (DG) operator, ES operator, and electricity consumers. The framework incorporates a multi-scenario stochastic model to address the uncertainties of DG output and load demand. By designing an adaptive optimization strategy based on reinforcement learning, the proposed method achieves coordinated optimization of electricity consumption among stakeholders, providing a novel solution to the challenges of multi-stakeholder distribution network planning. The proposed method is validated on the IEEE 33-bus and 69-bus extended systems. In the two test systems, the proposed method increased the total annual net benefits by 203k¥ and 116k¥ respectively. The results demonstrate that the method maximizes the economic benefits of all participants while significantly improving overall system performance, verifying its effectiveness and engineering applicability.

Index Terms—Distribution network, electricity consumption, reinforcement learning, collaborative planning, bi-level game.

I. INTRODUCTION

IN recent years, with the continuous growth of power load demand and the proposal of energy-saving and carbon reduction targets [1], [2], [3], traditional single-structure distribution networks have evolved into complex interactive systems integrating source, network, load, and storage (SNLS) [4], [5]. On the source side, the proportion of distributed generation (DG) integration—primarily photovoltaic (PV) and wind power—has significantly increased, promoting the development of a green economy [6]. On the network side, the ever-increasing electricity load compels the distribution system operator (DSO) to construct additional distribution lines to meet the connection demands of new nodes [7]. On the load side, electricity consumers actively adjust their electricity demand response to reduce peak demand [8]. On

the storage side, large-scale energy storage (ES) facilities are being deployed to achieve peak shaving and valley filling [9]. These four stakeholders have distinct interests in distribution network planning, leading to a complex multi-stakeholder gaming phenomenon [10]. Therefore, to achieve low-carbon goals while balancing the interests and operational coordination among stakeholders, a more flexible and robust distribution network planning method is imperative [11].

There are relatively more studies on distribution network planning from the perspective of a single planner. Prakash *et al.* [12] established a multi-objective optimization model with cost, productivity, power supply reliability, and other factors as objective functions to optimize system configuration. Zeng *et al.* [13] established a model for the siting and capacity determination of charging, switching and storage integrated station in AC-DC hybrid distribution network, which was solved by dynamic weights combined with binary genetic algorithm. Wang *et al.* [14] proposed an explicit method for optimal PV-storage capacity ratio in distribution network planning. By treating PV capacity as a parameter and converting the mixed-integer nonlinear model into a linear one, it analytically derives piecewise linear expressions for optimal storage capacity and total cost, avoiding the heavy computations of traditional methods. Zhang *et al.* [15] proposed a bi-level planning model for AC-DC hybrid distribution networks, which is solved using the normalization method towards the constraint method. Wu *et al.* [16] proposed an active distribution network SNLS bi-level collaborative planning model, which has the upper-level as the planning level and the lower-level as the operation level, and converts the model into a second-order cone programming problem. Pemmada *et al.* [17] proposed different heuristic algorithms for the site selection and capacity determination of distributed generation in distribution networks. Both methods significantly improve the operational performance of distribution networks while demonstrating good effectiveness and robustness.

In contrast, there is relatively little research literature on distribution network planning from the perspective of multiple stakeholders. Sun *et al.* [18] and Chen *et al.* [19] mainly studied the cooperative management problem of multiple integrated energy microgrids (IEM) under the same distribution network, and established a distribution network-multiple IEM cooperative optimization model based on a bi-level game from the perspective of the game between individual IEM members. Xiao *et al.* [20] considered the game-theoretic interactions among DSO, distribution companies, and distributed energy

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aggregators, with an iterative algorithm enabling information exchange between these entities. Lu *et al.* [21] investigated the transaction flow problem between multiple energy sources (electricity, gas, cooling, and heat), which was modeled as a bi-level game planning model between energy suppliers, integrated energy stations, and consumer subjects. Liu *et al.* [22] investigated bi-level game coordination planning for PV and ES integration in distribution networks, with the DSO modeled as the primary investment entity. Feng *et al.* [23] studied the cooperative planning problem of distribution network structure and DG, and established a master-slave game model based on the game between the grid company and the electricity consumers. Liu *et al.* [24] established a distribution network SNLS collaborative planning model based on master-slave games, which took into account the interactive relationships among four stakeholders.

Based on the aforementioned two research branches, we categorize existing distribution network planning studies into two main streams. First, many scholars adopt single-objective or multi-objective optimization models to carry out distribution network planning from the perspective of a single planner, and use genetic algorithm (GA), particle swarm optimization (PSO) and other meta-heuristic algorithms to solve the siting and sizing problems of DG and ES. Second, several studies introduce game theory to analyze the interactive relationship between partial market participants, and establish bi-level game models for limited stakeholders such as microgrids and grid companies. Despite the fruitful achievements of existing literature, three key research gaps still exist [25]:

- 1) Incomplete stakeholder modeling: Most game-based studies only consider 2–3 types of participants, and fail to fully couple long-term planning decisions and short-term operational strategies of DSO, DG operator, ES operator and power consumers simultaneously.
- 2) Defects of traditional solving methods: Meta-heuristic algorithms have been extensively applied to distribution network planning. Nevertheless, when addressing complex optimization problems characterized by high-dimensional state-action spaces, multi-stakeholder games and nonlinear constraints, these algorithms exhibit low computational efficiency, weak generalization ability and a tendency to converge to local optima easily.
- 3) Insufficient integration of uncertainty and game: Few studies embed the stochastic fluctuations of DG output and load demand into the bi-level game framework, resulting in poor robustness of planning schemes under actual variable operating conditions.

In contrast, emerging artificial intelligence technologies, particularly deep learning and reinforcement learning (RL), demonstrate strong optimization capabilities for high-dimensional nonlinear data with minimal dependence on manual expertise or mathematical models [26], [27], [28]. These advancements offer a promising pathway to overcome the technical bottlenecks in distribution network planning [29]. Li *et al.* [30] considered voltage deviation, power loss, statistical voltage stability, and generation cost as objective functions. The study employed DQN algorithm to address the distribution network reconfiguration problem. Liao *et al.* [31] proposed

a deep reinforcement learning framework for reactive power optimization in distribution networks with high penetration of distributed energy resources. Song *et al.* [32] proposed a distribution network line expansion planning method based on graph reinforcement learning (RL) and configured the sectionalizing switches.

To fill the above gaps, this paper takes the multi-stakeholder collaborative planning of SNLS distribution networks as the research object. We construct a bi-level game framework to characterize the interest interaction of all core participants, adopt a multi-scenario stochastic model to quantify external uncertainties, and further design a PPO-DDPG hybrid reinforcement learning algorithm to solve the complex mixed-variable game model. The main contributions are as follows.

- 1) The collaborative planning in the distribution network comprehensively considers four key participants: source, network, load, and storage. By establishing a mathematical model that couples the interests of multiple parties, it achieves full-chain coordinated optimization of the DG, DSO, electricity consumers, and ES.
- 2) The application of RL algorithm provides an effective solution for complex distribution network planning problems. The proposed framework constructs a state space incorporating multi-dimensional elements, which includes distribution network topology, node parameters, and electricity market prices. By designing an appropriate reward function mechanism, the agent can autonomously learn optimal planning strategies through environmental interactions.
- 3) A bi-level game model is established to capture the interactions between multiple agents. This model balances the interests of all participants in the SNLS system, effectively coordinates conflicting interests, and achieves intelligent allocation and dynamic balance of electricity consumption resources. This model provides key technical support for developing flexible, efficient, and low-carbon distribution networks, thus promoting power systems toward cleaner, smarter development.

This research aims to balance the interests of all stakeholders, improve the economic benefits and renewable energy consumption capacity of distribution networks, and overcome the limitations of traditional solving methods. The rest of this paper is organized as follows. Section II introduces the bi-level game model. Section III establishes detailed models for key stakeholders. Section IV introduces the RL algorithm for multi-agent. Section V verifies the rationality and effectiveness of the method through simulation cases. Section VI summarizes the work.

II. BI-LEVEL GAME-THEORETIC PLANNING MODEL

In the electricity market environment, various stakeholders participating in distribution network planning have distinct interests, leading to complex gaming scenarios due to their interactions and conflicts. This section establishes the game-theoretic framework using three fundamental elements of game theory and analyzes the equilibrium solution of the game. The interaction model is shown in Fig. 1.

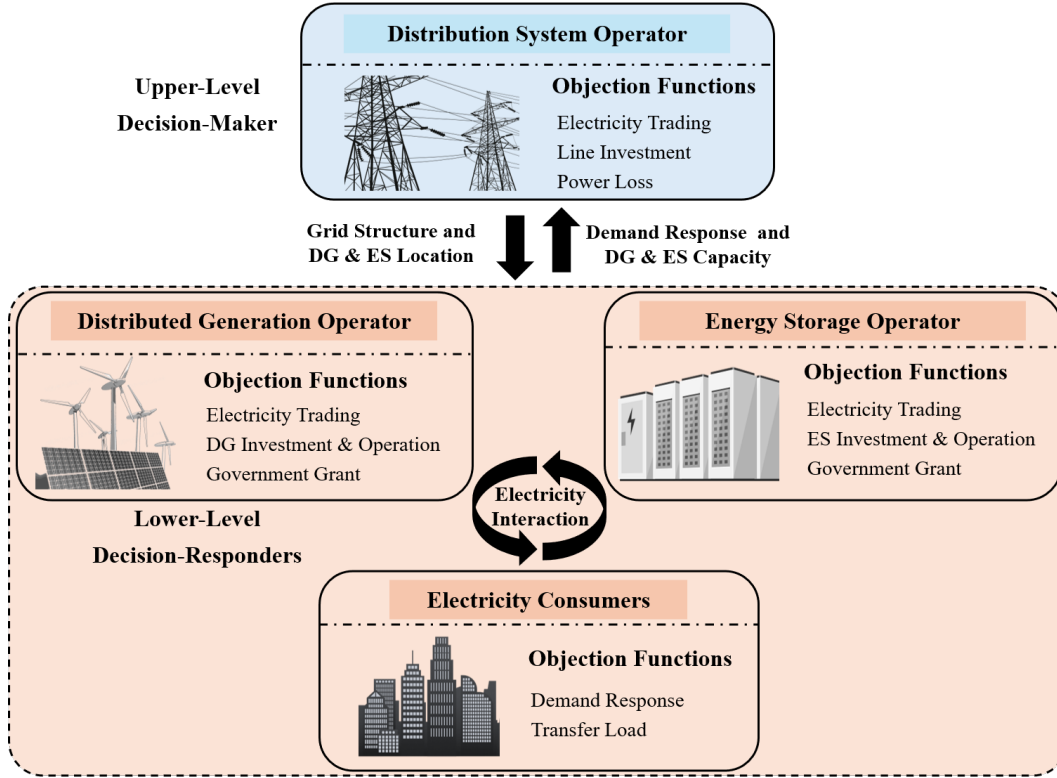


Fig. 1: Interaction model among stakeholders in bi-level game.

A. Game Theory Elements

1) *Players*: This paper organizes these four game participants into two interactive coalitions within a hierarchical game structure.

- Upper-level player
 - DSO: Purchase electricity from the upper-layer grid, supply end-consumers, expand/maintain the grid infrastructure to reduce power losses and ensure stable operation [33].
- Lower-level players
 - DG operator: Focus on the investment, construction, and maintenance of distributed renewable generation to increase renewable penetration, enhance system stability, and achieve emission reduction targets.
 - ES operator: Invest and operate storage facilities, optimize charging/discharging strategies based on temporal load variations to maximize profits while achieving peak-shaving and valley-filling objectives.
 - Electricity consumers: Adjust consumption patterns in response to price signals and incentive mechanisms [34].

2) *Strategies*: The DSO as the upper-level decision-maker, responsible for strategic infrastructure planning including distribution line routing and joint siting of DG and ES facilities. Simultaneously, the multi-stakeholder coalition (MSC) serves as the lower-level decision-responders. Based on the DSO's siting decisions, this coalition determines equipment capacity allocations and demand response strategies. The strategy spaces for each player are defined as:

- $\mathcal{S}_{\text{DSO}} = \{x_{ij}, y_k^{\text{DG}}, y_l^{\text{ES}}\}$, where x_{ij} represents line construction decisions, y_k^{DG} and y_l^{ES} denote DG and ES siting decisions respectively.
- $\mathcal{S}_{\text{MSC}} = \{p_k^{\text{DG}}, p_l^{\text{ES}}, p_m^{\text{DR}}\}$, where p_k^{DG} and p_l^{ES} represent capacity allocation decisions, and p_m^{DR} denotes demand response adjustments.

3) *Payoffs*: The payoff functions for each stakeholder are formulated as:

- DSO's payoff

$$\Pi_{\text{DSO}} = C_{\text{DSO}}, \quad (1)$$

where C_{DSO} denotes the annual consolidated revenue of the DSO, and it will be analyzed in section III (Eq. 4).

- MSC's payoff

$$\Pi_{\text{MSC}} = C_{\text{DG}} + C_{\text{ES}} + C_{\text{DR}}, \quad (2)$$

where C_{DG} and C_{ES} denote their respective annual consolidated revenue, and C_{DR} represents cost savings. All of them will be analyzed in section III (Eq. 11, 19, 29).

B. Game Formulation and Equilibrium Analysis

The interaction between players forms a bi-level hierarchical game that can be formally expressed as

$$\begin{cases} \max_{x,y} \Pi_{\text{DSO}}(x, y, p^*(x, y)), \\ \text{s.t. } (x, y) \in \mathcal{F}_{\text{DSO}}, \\ \text{where } p^*(x, y) = \arg \max_p \Pi_{\text{MSC}}(x, y, p), \\ \text{s.t. } p \in \mathcal{F}_{\text{MSC}}(x, y), \end{cases} \quad (3)$$

where $\mathcal{F}_{\text{DSO}}$ and $\mathcal{F}_{\text{MSC}}$ represent the feasible sets for DSO and MSC respectively.

The proposed bi-level game can obtain a unique equilibrium solution based on the following conditions (x^*, y^*, p^*) .

- 1) The strategy sets of both participants are nonempty compact convex sets.
- 2) For fixed (x^*, y^*) , p^* solves the lower-level MSC optimization problem.
- 3) (x^*, y^*) solves the upper-level DSO problem given MSC's optimal response $p^*(x, y)$.

All plays' constraints (introduced in section III) are linear constraints, and their payoff functions are nonempty compact convex sets. It can be proven that the equilibrium solution in the bi-level game is valid. Reasonable value ranges and hard constraints are set for all variables to prevent numerical singularity under diverse operating conditions. Meanwhile, all operational constraints are embedded into the model to ensure the feasibility of solutions.

III. MODELING REVENUES OF EVERY STAKEHOLDER

Based on the bi-level game framework introduced in Section II, this section establishes precise mathematical models for each stakeholder's revenue structure and related constraints.

A. Distribution System Operator Revenue Modeling

1) *Objective Function:* The DSO aims to maximize its annual comprehensive revenue, considering electricity trade profits, grid construction costs, and network loss expenses. The objective function is formulated as

$$\max C_{\text{DSO}} = C_{\text{DSO}}^{\text{net}} - C_{\text{DSO}}^{\text{invest}} - C_{\text{DSO}}^{\text{loss}}, \quad (4)$$

where $C_{\text{DSO}}^{\text{net}}$ represents the DSO's annual revenue from the sale and purchase of electricity, $C_{\text{DSO}}^{\text{invest}}$ denotes the cost of investment in new lines for DSO, and $C_{\text{DSO}}^{\text{loss}}$ is the annual network losses cost of the DSO.

The components (Eq. 4) are modeled as

$$C_{\text{DSO}}^{\text{net}} = \sum_{s=1}^{N_s} D_s \sum_{t=1}^T \left(\mu_{\text{DSO},t}^{\text{sell}} E_{\text{DSO},s,t}^{\text{sell}} - \mu_{\text{DSO},t}^{\text{buy}} E_{\text{DSO},s,t}^{\text{buy}} \right), \quad (5)$$

$$C_{\text{DSO}}^{\text{invest}} = \frac{b(1+b)^y}{(1+b)^y - 1} \sum_{i,j \in \Omega^{\text{line}}} \mu_{\text{line}} L_{i,j}, \quad (6)$$

$$C_{\text{DSO}}^{\text{loss}} = \sum_{s=1}^{N_s} D_s \sum_{t=1}^T \mu_t^{\text{loss}} E_{\text{DSO},s,t}^{\text{loss}}, \quad (7)$$

where $E_{\text{DSO},s,t}^{\text{sell}}$ (sales to consumers) and $E_{\text{DSO},s,t}^{\text{buy}}$ (purchases from upper-layer grid) are energy quantities at scenario s time t , with $\mu_{\text{DSO},t}^{\text{sell}}$ and $\mu_{\text{DSO},t}^{\text{buy}}$ as corresponding prices, $E_{\text{DSO},s,t}^{\text{loss}}$ represents active power loss at time t , priced at μ_t^{loss} . N_s is scenario count, D_s is duration per scenario (days), T is the number of hours in a scene, Ω^{line} is the set of lines, μ_{line} is unit line cost, $L_{i,j}$ is length between nodes i, j , b is discount rate, and y is equipment lifespan (years).

2) *Restrictive Conditions:* The distribution network operation must satisfy three fundamental physical constraints: the power flow balance equations (Eq. 8), nodal voltage operating limits (Eq. 9), and branch thermal capacity constraints (Eq. 10). These constraints are modeled as

$$\begin{cases} P_{ij} = U_i \sum_{j \in i} U_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}), \\ Q_{ij} = U_i \sum_{j \in i} U_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}), \end{cases} \quad (8)$$

$$U_{i,\min} \leq U_i \leq U_{i,\max}, \quad (9)$$

$$\sqrt{P_{ij}^2 + Q_{ij}^2} \leq S_{ij,\max}, \quad (10)$$

where P_{ij}, Q_{ij} denote active/reactive branch flows, U_i, U_j are nodal voltages, $G_{ij} + jB_{ij}$ is the branch admittance, and θ_{ij} is the voltage angle difference. $U_{i,\min}$ and $U_{i,\max}$ are distribution system min/max voltage, $S_{ij,\max}$ is the apparent power limit for branch ij determined by conductor ratings.

B. Distributed Generation Operator Revenue Modeling

1) *Objective Function:* The DG operator aims to maximize its annual comprehensive revenue, considering electricity sales income, government subsidies, construction investments, and operational costs. The objective function is formulated as

$$\max C_{\text{DG}} = C_{\text{DG}}^{\text{sell}} + C_{\text{DG}}^{\text{grant}} - C_{\text{DG}}^{\text{invest}} - C_{\text{DG}}^{\text{operate}}, \quad (11)$$

where $C_{\text{DG}}^{\text{sell}}$ denotes the electricity sales revenue, $C_{\text{DG}}^{\text{grant}}$ is the government subsidies, $C_{\text{DG}}^{\text{invest}}$ represents the construction costs, and $C_{\text{DG}}^{\text{operate}}$ is the operational expenses.

The components (Eq. 11) are modeled as

$$C_{\text{DG}}^{\text{sell}} = \sum_{s=1}^{N_s} D_s \sum_{t=1}^T \sum_{i \in \Omega^{\text{DG}}} \mu_{\text{DG},t}^{\text{sell}} E_{i,s,t}^{\text{DG}}, \quad (12)$$

$$C_{\text{DG}}^{\text{grant}} = \sum_{s=1}^{N_s} D_s \sum_{t=1}^T \sum_{i \in \Omega^{\text{DG}}} \mu_{\text{DG},t}^{\text{grant}} E_{i,s,t}^{\text{DG}}, \quad (13)$$

$$C_{\text{DG}}^{\text{invest}} = \frac{b(1+b)^y}{(1+b)^y - 1} \sum_{i \in \Omega^{\text{DG}}} \mu_{\text{DG}}^{\text{invest}} S_i^{\text{DG}}, \quad (14)$$

$$C_{\text{DG}}^{\text{operate}} = \sum_{s=1}^{N_s} D_s \sum_{t=1}^T \sum_{i \in \Omega^{\text{DG}}} \mu_{\text{DG},t}^{\text{operate}} E_{i,s,t}^{\text{DG}}, \quad (15)$$

where $E_{i,s,t}^{\text{DG}}$ represents power output at node i (scenario s , time t), Ω^{DG} denotes the set of DG nodes, $\mu_{\text{DG},t}^{\text{sell}}$ is the electricity price, $\mu_{\text{DG},t}^{\text{grant}}$ is the subsidy rate, $\mu_{\text{DG}}^{\text{invest}}$ is unit capacity cost, $\mu_{\text{DG},t}^{\text{operate}}$ is the operational cost coefficient, and S_i^{DG} is the installed capacity.

2) *Restrictive Conditions*: The DG operation must satisfy three fundamental physical constraints: the capacity limits (Eq. 16), price limits (Eq. 17) and output power limits (Eq. 18). These constraints are modeled as

$$S_{i,\min}^{\text{DG}} \leq S_i^{\text{DG}} \leq S_{i,\max}^{\text{DG}}, \quad (16)$$

$$\mu_{\text{DG},\min}^{\text{sell}} \leq \mu_{\text{DG},t}^{\text{sell}} \leq \mu_{\text{DG},\max}^{\text{sell}}, \quad (17)$$

$$\begin{cases} P_{i,\min}^{\text{DG}} \leq P_{i,s,t}^{\text{DG}} \leq P_{i,\max}^{\text{DG}} \\ Q_{i,\min}^{\text{DG}} \leq Q_{i,s,t}^{\text{DG}} \leq Q_{i,\max}^{\text{DG}} \end{cases} \quad (18)$$

where $S_{i,\min}^{\text{DG}}$ and $S_{i,\max}^{\text{DG}}$ denote min/max installable capacities, $\mu_{\text{DG},\min}^{\text{sell}}$ and $\mu_{\text{DG},\max}^{\text{sell}}$ represent min/max sell prices. $P_{i,s,t}^{\text{DG}}$ and $Q_{i,s,t}^{\text{DG}}$ are active/reactive power outputs, with min/max superscripts denoting operational limits.

C. Energy Storage Operator Revenue Modeling

1) *Objective Function*: The ES operator aims to maximize their own revenue for the capacity allocation of ES equipment, taking into account their own revenue from the purchase and sale of electricity, state subsidies, ES construction and operation. The objective function is formulated as

$$\max C_{\text{ES}} = C_{\text{ES}}^{\text{net}} + C_{\text{ES}}^{\text{grant}} - C_{\text{ES}}^{\text{invest}} - C_{\text{ES}}^{\text{operate}}, \quad (19)$$

where $C_{\text{ES}}^{\text{net}}$ denotes the electricity arbitrage revenue, $C_{\text{ES}}^{\text{grant}}$ is the government subsidies, $C_{\text{ES}}^{\text{invest}}$ is the construction costs, and $C_{\text{ES}}^{\text{operate}}$ is the operational expenses.

The components (Eq. 19) are modeled as

$$C_{\text{ES}}^{\text{net}} = \sum_{s=1}^{N_s} D_s \sum_{t=1}^T \sum_{i \in \Omega^{\text{ES}}} \left(\mu_{\text{ES},t}^{\text{sell}} E_{i,s,t}^{\text{ES,sell}} - \mu_{\text{ES},t}^{\text{buy}} E_{i,s,t}^{\text{ES,buy}} \right), \quad (20)$$

$$C_{\text{ES}}^{\text{grant}} = \sum_{s=1}^{N_s} D_s \sum_{t=1}^T \sum_{i \in \Omega^{\text{ES}}} \mu_{\text{ES},t}^{\text{grant}} E_{i,s,t}^{\text{ES,sell}}, \quad (21)$$

$$C_{\text{ES}}^{\text{invest}} = \frac{b(1+b)^y}{(1+b)^y - 1} \sum_{i \in \Omega^{\text{ES}}} \mu_{\text{ES}}^{\text{invest}} S_i^{\text{ES}}, \quad (22)$$

$$C_{\text{ES}}^{\text{operate}} = \sum_{s=1}^{N_s} D_s \sum_{t=1}^T \sum_{i \in \Omega^{\text{ES}}} \mu_{\text{ES},t}^{\text{operate}} E_{i,s,t}^{\text{ES,sell}}, \quad (23)$$

where $E_{i,s,t}^{\text{ES,sell}}$ (discharge) and $E_{i,s,t}^{\text{ES,buy}}$ (charge) are energy quantities at node i (scenario s , time t), with $\mu_{\text{ES},t}^{\text{sell}}$ and $\mu_{\text{ES},t}^{\text{buy}}$ as corresponding prices. $\mu_{\text{ES},t}^{\text{grant}}$ is the subsidy rate, $\mu_{\text{ES}}^{\text{invest}}$ denotes unit capacity cost, $\mu_{\text{ES},t}^{\text{operate}}$ is the operational cost coefficient, and S_i^{ES} is the installed capacity.

2) *Restrictive Conditions*: The ES operation must satisfy the following fundamental physical constraints: the capacity limits (Eq. 24), price limits (Eq. 25) and operational limits (Eq. 26-28). These constraints are modeled as

$$S_{i,\min}^{\text{ES}} \leq S_i^{\text{ES}} \leq S_{i,\max}^{\text{ES}}, \quad (24)$$

$$\mu_{\text{ES},\min}^{\text{sell}} \leq \mu_{\text{ES},t}^{\text{sell}} \leq \mu_{\text{ES},\max}^{\text{sell}}, \quad (25)$$

$$0 \leq P_{i,s,t}^{\text{ES,dis}} \leq \alpha_{i,s,t}^{\text{ES}} P_i^{\text{ES}}, \quad (26)$$

$$E_{i,s,t+\Delta t}^{\text{ES}} = E_{i,s,t}^{\text{ES}} + \left(\beta_i^{\text{ES,ch}} P_{i,s,t}^{\text{ES,ch}} - \frac{P_{i,s,t}^{\text{ES,dis}}}{\beta_i^{\text{ES,dis}}} \right), \quad (27)$$

$$E_{i,\min}^{\text{ES}} \leq E_{i,s,t}^{\text{ES}} \leq E_{i,\max}^{\text{ES}}, \quad (28)$$

where $\alpha_{i,s,t}^{\text{ES}} \in \{0, 1\}$ is the discharge flag, $E_{i,s,t}^{\text{ES}}$ is the current stored electricity at time t , $\beta_i^{\text{ES,ch/dis}}$ is the charge/discharge efficiency, and Δt is the time slot duration. $E_{i,\min}^{\text{ES}}$ and $E_{i,\max}^{\text{ES}}$ denote min/max allowable electricity.

D. Electricity Consumers Revenue Model

1) *Objective Function*: Electricity consumers aim to reduce electricity costs by maximizing demand response programs. The objective function is formulated as

$$\max C_{\text{DR}} = \sum_{s=1}^{N_s} D_s \sum_{t=1}^T \sum_{i \in \Omega^{\text{DR}}} \mu_{\text{DR}} \left(E_{i,s,t}^{\text{DR,out}} - E_{i,s,t}^{\text{DR,in}} \right), \quad (29)$$

where μ_{DR} denotes the incentive price, $E_{i,s,t}^{\text{DR,out}}$ and $E_{i,s,t}^{\text{DR,in}}$ are the load reduction/increase at node i .

2) *Restrictive Conditions*: The DR must satisfy the following fundamental physical constraints: the load transfer limits (Eq. 30-31) and the energy balance (Eq. 32). These constraints are modeled as

$$E_{i,\min}^{\text{DR,out}} \leq E_{i,s,t}^{\text{DR,out}} \leq E_{i,\max}^{\text{DR,out}}, \quad (30)$$

$$E_{i,\min}^{\text{DR,in}} \leq E_{i,s,t}^{\text{DR,in}} \leq E_{i,\max}^{\text{DR,in}}, \quad (31)$$

$$\sum_{t=1}^T \left(E_{i,s,t}^{\text{DR,out}} - E_{i,s,t}^{\text{DR,in}} \right) = 0, \quad (32)$$

where all min/max superscripts denote operational boundaries for load shifting.

IV. PPO-DDPG MULTI-AGENT REINFORCEMENT LEARNING ALGORITHM DESIGN

RL is a feedback-driven method where an agent interacts with an environment by taking actions, receiving rewards, and transitioning between states. Through trial and error, the agent optimizes its policy based on reward signals. Such characteristics make RL well-suited for complex decision-making problems like distribution network planning and active voltage control [35], [36].

This section proposes a hybrid multi-agent RL framework that combines proximal policy optimization (PPO) and deep deterministic policy gradient (DDPG), which aims to address the following key issues: (1) how to construct a Markov Decision Process (MDP) framework suitable for multi-agent gaming; (2) how to design policy interactions among multiple agents; (3) how to achieve effective policy learning in high-dimensional state spaces.

A. Construction of Markov Decision Process

A MDP is defined as a tuple (S, A, P, R, γ) [37], where:

- S is a finite set of states
- A is a finite set of actions
- $P : S \times A \times S \rightarrow [0, 1]$ represents the state transition probability
- $R : S \times A \rightarrow \mathbb{R}$ denotes the reward function
- $\gamma \in [0, 1]$ is the discount factor

The agent's action selection policy $\pi : S \rightarrow A$ depends only on the current state and satisfies the Markov property

$$\Pr(s_{t+1}|s_t, a_t) = \Pr(s_{t+1}|s_t, a_t, s_{t-1}, \dots, s_0). \quad (33)$$

In the distribution network planning framework, the MDP initiates from an initial planning scheme $s_0 \in S$ and proceeds through iterative optimization cycles. At each time step t , the current network configuration s_t undergoes modification to generate an improved scheme s_{t+1} . This iterative process continues until achieving convergence, typically when the reward difference between consecutive iterations satisfies $\|R_{t+1} - R_t\| < \epsilon$, where ϵ represents a predefined tolerance threshold. Upon convergence, the environment resets to commence subsequent training episodes, which ensures comprehensive exploration of the planning solution space.

The proposed bi-level game framework comprises two interacting decision-making agents: the upper-level agent represented by the DSO responsible for grid infrastructure planning, and the lower-level agent consisting of a MSC that determines equipment operation strategies.

Both agents share the state space $S = (B, C, V, R)$ where

$$B \in \{0, 1\}^n, \quad C \in \mathbb{R}^n, \quad V \in \mathbb{R}^n, \quad R \in \{0, 1\}^m, \quad (34)$$

with:

- B : Node installation matrix (1=installed, 0=not installed)
- C : Equipment capacity matrix
- V : Node voltage magnitude matrix
- R : Line construction matrix

The state space is updated in real time to cope with sudden load variations, enabling rapid adjustment of agent strategies. The action spaces follow a hierarchical structure with clearly defined decision domains. The upper-level action space (A_{DSO}) encompasses geographical planning decisions including DG installation sites, ES equipment locations, and new distribution circuits. The lower-level action space (A_{MSC}) addresses capacity allocation parameters and demand response, specifically determining the installed capacities for DG and ES equipment along with corresponding electricity prices.

The reward functions are defined as

$$R_{\text{DSO}} = \Pi_{\text{DSO}}, \quad (35)$$

$$R_{\text{MSC}} = \Pi_{\text{MSC}}. \quad (36)$$

B. PPO-DDPG Hybrid Reinforcement Learning Algorithm

This paper adopts a bi-level game framework combined with a hybrid PPO-DDPG reinforcement learning algorithm to solve the multi-stakeholder collaborative planning problem of distribution networks. The bi-level game naturally describes

the master-slave decision-making relationship between the distribution system operator and lower-level stakeholders. The PPO-DDPG framework decouples discrete planning variables and continuous operational variables, effectively handling the mixed-integer non-convex characteristics of the problem. Compared with traditional meta-heuristic and convex optimization methods, the proposed method achieves better convergence performance, stronger robustness against uncertainties, and higher real-time inference speed, making it suitable for both offline planning and online dynamic optimization.

In the bi-level game-theoretic framework, the upper-level DSO determines the overall grid topology configuration of the power distribution system, including the layout of new distribution circuits as well as the installation locations for DG and ES equipment. To solve this discrete decision-making problem, our study employs the PPO algorithm [38]. As an on-policy RL method based on policy gradients, PPO demonstrates superior training stability, broad applicability, and excellent computational efficiency.

The lower-level MSC is responsible for determining both the capacity allocation of newly installed equipment and dynamic electricity pricing adjustments, which are jointly optimized using the DDPG algorithm. Specifically designed for continuous action spaces, DDPG is particularly well-suited for addressing continuous decision-making problems such as capacity configuration. This algorithm adopts an Actor-Critic architecture and operates as an off-policy method.

1) *PPO Algorithm*: PPO is a policy optimization-based RL method that employs stochastic policies to handle problems with continuous state and action spaces. By constraining the magnitude of policy updates, PPO ensures each training step maintains close proximity to the current policy while efficiently utilizing sampled data. The algorithm evaluates the relative quality of actions through an advantage function estimation mechanism.

The objective function is defined as

$$L^{\text{clip}}(\theta) = \arg \max_{\theta} \mathbb{E}_{s \sim \rho_{\theta_k}} \mathbb{E}_{a \sim \pi_{\theta_k}(\cdot|s)} \left[\text{clip} \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_k}(a|s)}, 1 - \epsilon, 1 + \epsilon \right) A^{\pi_{\theta_k}}(s, a) \right], \quad (37)$$

where:

- The probability ratio $r_t(\theta) = \frac{\pi_{\theta}(a|s)}{\pi_{\theta_k}(a|s)}$ compares the action selection probabilities between new and old policies
- $\hat{A}_t = A^{\pi_{\theta_k}}(s, a)$ represents the advantage function estimate
- The clip operation $\text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)$ constrains the ratio within $(1 - \epsilon, 1 + \epsilon)$

This clipped objective achieves a crucial balance. It enables policy improvement while preventing excessively large updates that could destabilize training. The hyperparameter ϵ (typically 0.1-0.3) controls the maximum allowed policy change per iteration.

The PPO algorithm implements optimization through the following key steps:

1. *Advantage Function Calculation*: Evaluate the relative quality of action a_t in state s_t compared to the average

performance (advantage $\hat{A}_t$)

$$\hat{A}_t^{GAE} = \sum_{l=0}^{T-t+1} (\gamma\lambda)^l \delta_{t+l}, \quad (38)$$

$$\delta_{t+l} = r_{t+l} + \gamma V(s_{t+l}) - V(s_t), \quad (39)$$

where $\gamma \in (0.9, 0.99)$ is the discount factor and $\lambda \in (0.9, 1.0)$ controls the bias-variance tradeoff.

2. Probability Ratio Computation: Compare action selection probabilities between new and old policies

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}. \quad (40)$$

3. Policy Update: Apply clipping to penalize excessive policy updates beyond the range $(1 - \epsilon, 1 + \epsilon)$, which ensures moderate update magnitudes.

4. Value Function Update: Perform value function regression using multi-step TD targets

$$L^{VF}(\theta) = \frac{1}{2} \mathbb{E}_t \left[(\hat{A}_t + V_\theta(s_t) - V_{target}(s_t))^2 \right]. \quad (41)$$

2) *DDPG Algorithm*: The DDPG algorithm represents an off-policy, model-free RL method within the actor-critic framework, specifically optimized for continuous control problems [39]. In contrast to stochastic policy methods such as PPO, DDPG utilizes a deterministic policy that directly maps states to precise action values. This characteristic makes it particularly suitable for continuous control tasks in distribution network planning, such as generator output regulation and ES charge/discharge control.

The algorithm employs four neural networks in a dual-network configuration:

- Online Actor Network ($\mu(s|\theta^\mu)$): Execute state-to-action mapping for policy evaluation
- Online Critic Network ($Q(s, a|\theta^Q)$): Estimate state-action values during policy evaluation
- Target Actor Network ($\mu'(s'|\theta^{\mu'})$): Provide delayed policy evaluation for stability
- Target Critic Network ($Q'(s', a'|\theta^{Q'})$): Generate stable Q-value targets for policy improvement

The algorithm implements optimization through the following key steps:

1. Value Function Estimation: The critic update utilizes the modified Bellman Operator

$$\mathcal{T}^\mu Q(s, a) = \mathbb{E} [r + \gamma Q'(s', \mu'(s'))], \quad (42)$$

with the loss function

$$\mathcal{L}_Q(\theta^Q) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[(Q(s, a) - y(r, s'))^2 \right], \quad (43)$$

where $y(r, s') = r + \gamma Q'(s', \mu'(s'))$.

2. Policy Improvement: The actor update follows the deterministic policy gradient

$$\nabla_{\theta^\mu} J = \mathbb{E}_{s \sim \mathcal{D}} \left[\nabla_a Q(s, a|\theta^Q)|_{a=\mu(s)} \nabla_{\theta^\mu} \mu(s|\theta^\mu) \right]. \quad (44)$$

3. Target Network Synchronization: The target networks employ Polyak averaging

$$\theta^{Q'} \leftarrow \tau \theta^Q + (1 - \tau) \theta^{Q'}, \quad (45)$$

$$\theta^{\mu'} \leftarrow \tau \theta^\mu + (1 - \tau) \theta^{\mu'}, \quad (46)$$

with $\tau \in (0, 1)$ typically set to 0.001 for stable learning.

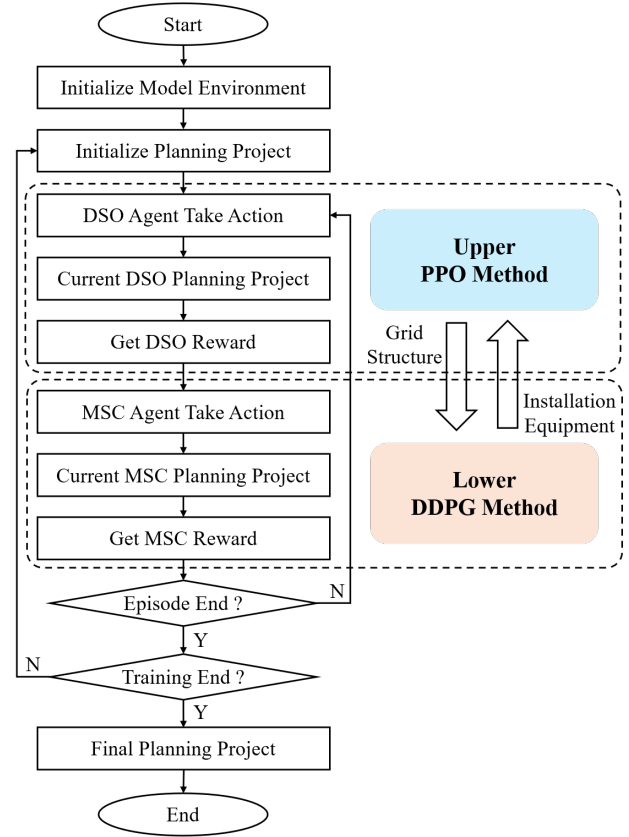


Fig. 2: PPO-DDPG algorithm process of distribution network planning.

C. PPO-DDPG Algorithm Training Process

Before the RL training, the whole model environment and planning scheme are initialized. The upper-level DSO agent first trains a grid architecture planning scheme based on the current initialization scheme and passes it to the lower-level MSC agent. The MSC agent trains the optimal installation equipment scheme for this architecture and feeds the results back to the upper-level DSO agent. These two agents play the game repeatedly with the goal of maximizing their respective rewards and update their planning schemes until the iteration converges. The complete execution flow of the proposed hybrid reinforcement learning algorithm is shown in Fig. 2. It includes the initialization phase, the iterative game process between the upper-level and lower-level agents, the policy update step, and the convergence criterion, which clearly illustrates how the proposed method solves the bi-level game planning problem. The proposed method adopts hybrid offline training and online inference. The offline training requires iterative game interaction and neural network optimization, while online inference achieves high computational efficiency, which satisfies the requirements of practical day-ahead distribution network planning.

V. SIMULATION STUDIES

To validate the proposed distribution network planning model, simulations are performed on IEEE 33-bus and 69-bus

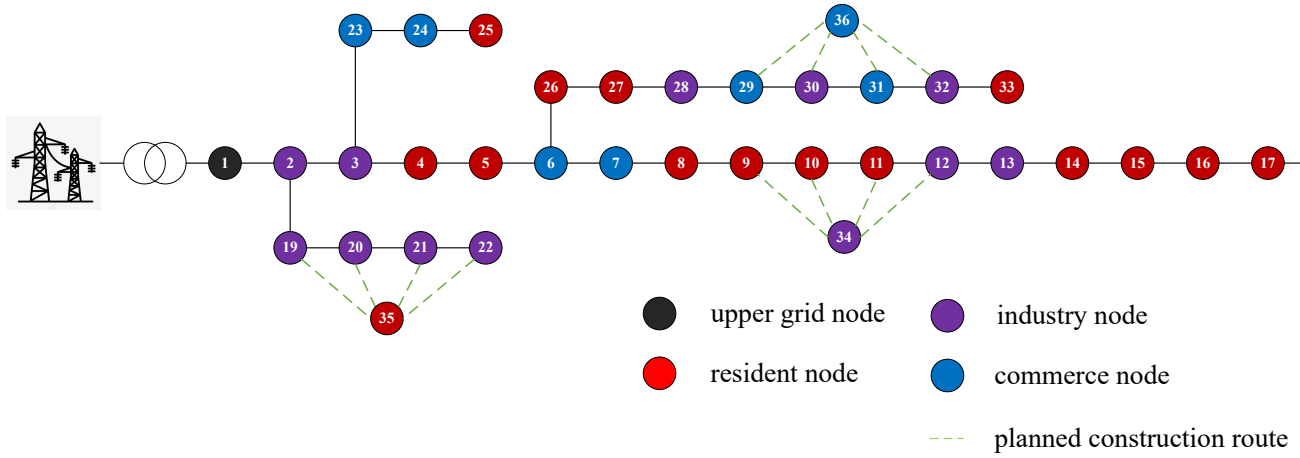


Fig. 3: IEEE 33-bus initial topology.

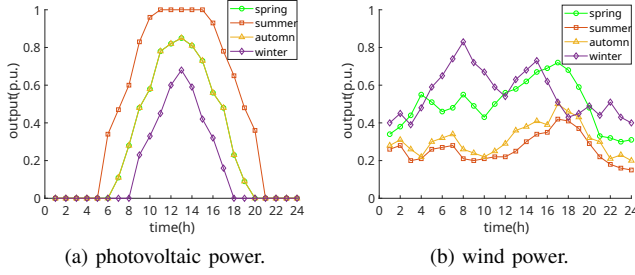


Fig. 4: DG output of different typical day.

TABLE I: RELATED PARAMETERS

Type	Parameter	Value
Wind Power	Max installation capacity (kW)	1000
	Investment cost (¥/kW)	5500
	Operation cost (¥/kWh)	0.3
Photovoltaic	Max installation capacity (kW)	1000
	Investment cost (¥/kW)	4500
	Operation cost (¥/kWh)	0.2
Energy Storage	Max installation capacity	1000 kW/800 kWh
	Investment cost (¥/kWh)	1000
	Operation cost (¥/kWh)	0.2
	SOC upper/lower limit	0.9/0.1
	Charge/discharge efficiency	0.9

TABLE II: TIME-OF-USE ELECTRICITY PRICE

Time Period	Price (¥/kWh)
Trough 0:00-8:00	0.37
Normal 8:00-12:00, 17:00-19:00, 22:00-24:00	0.68
Peak 12:00-17:00, 19:00-22:00	0.96

extended systems. All numerical experiments are performed on a computational platform with the following specifications:

- Hardware: Intel Core i5-8300H CPU @ 2.30GHz, 16GB RAM, NVIDIA GTX 1050Ti GPU
- Software: Python 3.7 with PyTorch 1.8.1 for PPO-DDPG algorithm implementation
- Power flow solver: PyPower 7.1 for network calculations

A. IEEE 33-Bus Extended System

The case study is based on an IEEE 33-bus extended system, with the initial topology of the distribution network shown in Fig. 3. To accommodate the growing load demand, new load nodes 34–36 are added to the distribution network, resulting in a maximum system load of 4015 kW + j2460 kVar. These load nodes are categorized into residential, industrial, and commercial loads, each with distinct power consumption characteristics. The initial IEEE 33-bus extended system is configured with PV, wind power, and ES equipment, as well as new distribution circuits. The parameters of various types of equipment are shown in Table I. The year is divided into four typical day types, and the PV and wind power output curves for each typical day are shown in Fig. 4. The time-of-use electricity purchase prices from the upper-layer grid for the entire distribution system are shown in Table II. All consumer-side loads are transferable and participate in demand response, shifting peak period electricity consumption to through period.

To validate the effectiveness of the proposed RL and bi-level game-based distribution network SNLS planning method, three methods are set up for solution:

- Method 1: Employ the PPO-DDPG RL algorithm for multi-stakeholder bi-level game-based distribution network SNLS coordination planning (the proposed method in this paper).
- Method 2: Use a heuristic algorithm for multi-stakeholder bi-level game-based distribution network SNLS coordination planning.
 - 2-1: PSO-GA algorithm.
 - 2-2: PSO algorithm.
 - 2-3: GA algorithm.
- Method 3: Exclude multi-stakeholder bi-level game-based distribution network SNLS coordination planning.

Comparative analysis between Methods 1-3 demonstrates the necessity of game-theoretic mechanisms, while Methods 1-2 comparison validates the RL advantage.

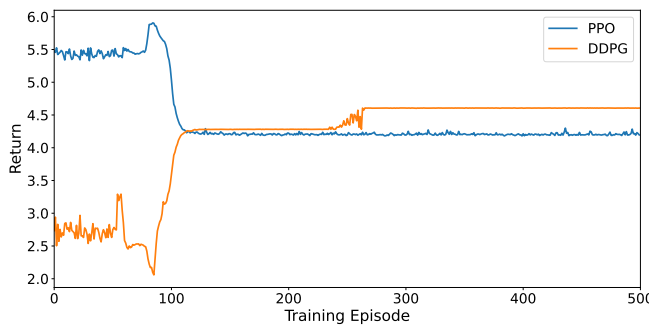


Fig. 5: IEEE 33-bus system PPO-DDPG (Method 1) training returns.

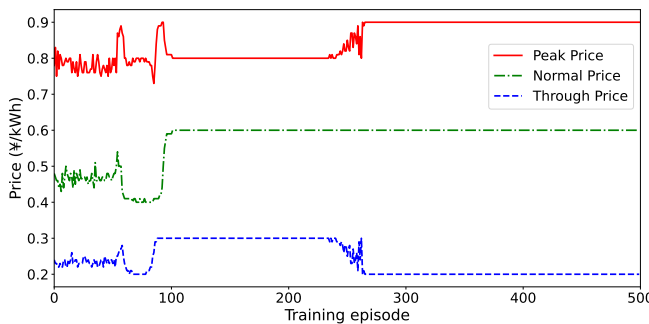


Fig. 6: IEEE 33-bus system PPO-DDPG (Method 1) electricity price changes.

Method 1 uses the DDPG-PPO algorithm, with the training process shown in Fig. 5. The changes in electricity prices are shown in Fig. 6. The 24 hours of a day are divided into three time periods: peak, trough, and normal. A total of 500 training episodes are conducted, and the planning scheme converges. During the first approximately 70 training episodes, the reward functions of the two agents fluctuate within a small range, reflecting the data collection process in RL. Subsequently, the reward function of the PPO agent shows a decreasing trend, while that of the DDPG agent shows an increasing trend. This is because when constructing DG and ES facilities, the increase in power generation inevitably reduces the amount of electricity purchased by consumers from the DSO, thereby decreasing the DSO's revenue while increasing the MSC's revenue. After more than 100 training episodes, the reward functions of the two agents stabilize.

Table III presents the results of different methods for planning location, capacity (L&C), and line routes. The PSO-GA algorithm performed best in Method 2. The following analysis refers to Method 2 as this algorithm. In Methods 1 and 2, DG operator maximizes the grid-connected capacity because increasing capacity boosts their electricity sales and profits when prioritizing their own interests. Method 3 results in smaller grid-connected capacity as it ignores individual stakeholder interests. Instead, it prioritizes overall benefits and makes upper-grid electricity purchases more profitable. In terms of ES capacity, Methods 1 and 2 also have the maximum grid-connected capacity. This is because ES operator can directly trade within the MSC to purchase electricity at a lower price and sell it during peak consumption periods. In

TABLE III: PLANNING RESULTS OF DIFFERENT METHODS

Method	DG L&C(MW)	ES L&C(MW)	Line construction
1	21(1) 18(1), 16(1)	22(1)	9-34 20-35,30-36
2-1	16(1) 17(1), 18(1)	15(1)	9-34 20-35,30-36
2-2	16(1) 21(1), 18(1)	12(1)	10-34 19-35,29-36
2-3	21(1) 22(1), 20(1)	19(1)	10-34 19-35,30-36
3	20(0.8) 21(0.8), 22(0.8)	2(0.8)	9-34 19-35,30-36

TABLE IV: PLANNING BENEFITS FOR STAKEHOLDERS

Method	Benefits/(10k¥)				
	C_{DSO}	C_{DG}	C_{ES}	C_{DR}	C_{total}
1	424.3	344.2	7.5	109.3	885.3
2-1	429.4	319.1	7.4	109.1	865
2-2	429.3	318.9	7.2	109.2	864.6
2-3	426.7	318.8	7.3	109.1	861.9
3	599.1	220.1	5.4	27.3	851.9

terms of node selection, all three methods focus on installing DG and ES at the end of the branch line. The reason for this is that line impedance causes voltage to drop as distribution distance increases in distribution network. Remote nodes typically exhibit lower voltage levels due to their greater distance from power generation nodes. Installing DG and ES can compensate for load current through local power generation, thereby reducing power flow on the lines and improving voltage quality. Additionally, the installation of DG and ES can shorten power flow paths, reducing line losses caused by high current distribution, particularly in long lines with high resistance, where this effect is more pronounced. This effect is more pronounced in long lines with high resistance. In the selection of new lines, both Method 1 and Method 2 can select Lines 9-34, 20-35, and 30-36. The reason is that these lines have the shortest lengths, maximizing cost savings for the DSO in terms of construction expenses.

Table IV shows the annual comprehensive benefits for each stakeholder group via different methods. From the table, it can be seen that the method using a bi-level game results in lower benefits for the DSO compared to Method 3, but the benefits for other groups are higher than those of Method 3. This is due to the bargaining power possessed by the lower-level stakeholders, enabling them to achieve higher benefits. Methods 1 and 2 have the same demand response benefits, as the demand response has reached the maximum benefit value after applying the bi-level game. Method 1 employs a RL algorithm, which leverages complex neural network fitting to achieve greater benefits in price response than Method 2's heuristic algorithm. It has been determined that Method 1 is the most profitable of the methods under consideration. This is due to the fact that the DG and energy storage benefits offered by Method 1 are superior to those of Method 2. Based on the

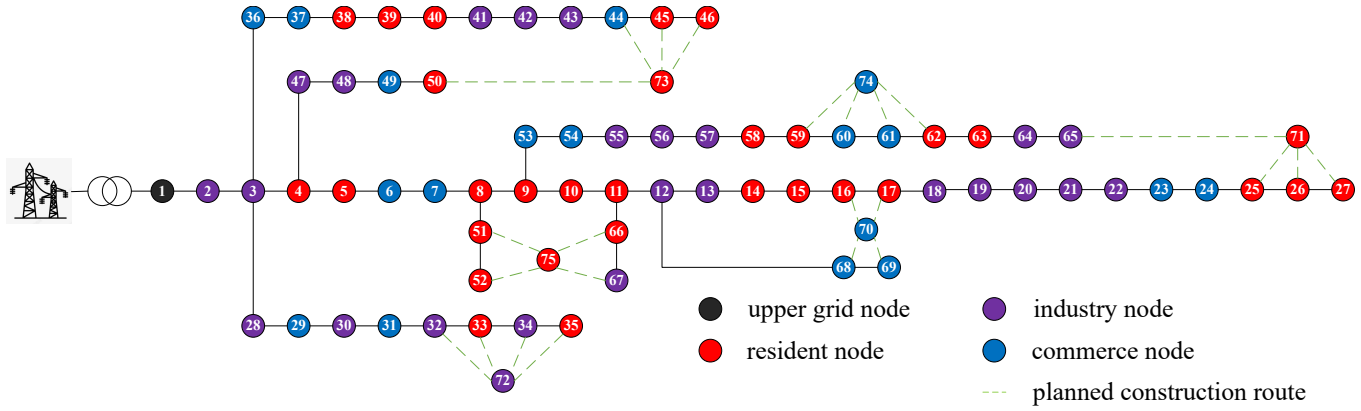


Fig. 7: IEEE 69-bus initial topology.

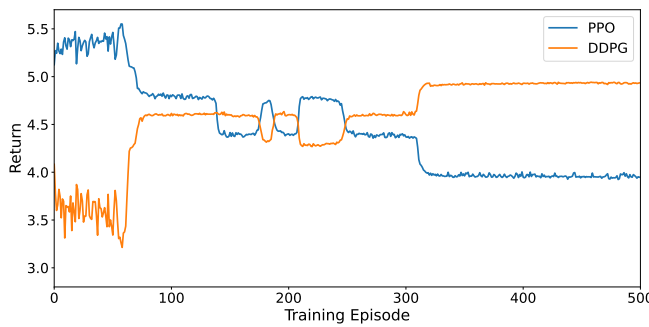


Fig. 8: IEEE 69-bus system PPO-DDPG (Method 1) training returns.

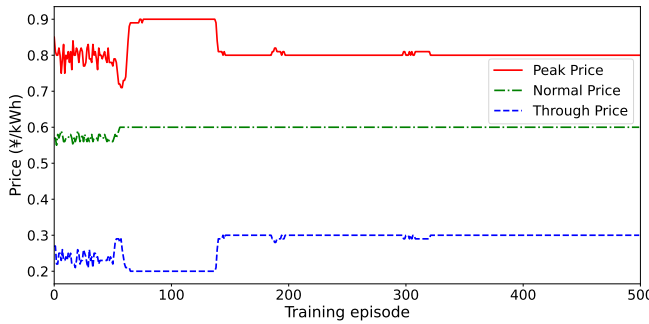


Fig. 9: IEEE 69-bus system PPO-DDPG (Method 1) electricity price changes.

above analysis, it can be concluded that increasing the grid-connected capacity of distributed energy after planning helps reduce the distribution grid's dependence on the upper-layer grid under the influence of a bi-level game.

B. IEEE 69-Bus Extended System

To validate the applicability of the proposed method in distribution network systems, a larger-scale case based on the IEEE 69-bus extended system is set up. The initial topology of the distribution network is shown in Fig. 7. New load nodes 70–75 are added to the distribution network, resulting in a maximum system load of 4176 kW + j2954 kVar. The maximum capacity of the DG and ES are adjusted to 700

TABLE V: PLANNING RESULTS OF DIFFERENT METHODS

Method	DG L&C(MW)	ES L&C(MW)	Line construction
1	35(0.7), 41(0.7) 38(0.7) 34(0.7), 9(0.7)	8(0.7), 24(0.7)	68–70, 26–71, 32–72, 46–73, 61–74, 66–75
2-1	45(0.7), 33(0.7) 46(0.7) 35(0.7), 34(0.7)	6(0.7), 29(0.7)	68–70, 65–71, 33–72, 50–73, 62–74, 67–75
2-2	26(0.7), 27(0.7) 25(0.7) 46(0.7), 54(0.7)	19(0.7), 30(0.7)	68–70, 65–71, 32–72, 50–73, 62–74, 67–75
2-3	46(0.7), 31(0.7) 35(0.7) 34(0.7), 33(0.7)	8(0.7), 30(0.7)	68–70, 26–71, 32–72, 45–73, 60–74, 67–75
3	29(0.5), 34(0.5) 35(0.5) 45(0.5), 41(0.5)	43(0.7), 33(0.67)	68–70, 27–71, 34–72, 50–73, 59–74, 66–75

TABLE VI: PLANNING BENEFITS FOR STAKEHOLDERS

Method	Benefits/(10k¥)				
	C_{DSO}	C_{DG}	C_{ES}	C_{DR}	C_{total}
1	399.1	364.8	13.6	115.5	893.0
2-1	403.5	349.5	13.1	115.3	881.4
2-2	401.6	349.2	13.4	115.2	879.4
2-3	404.0	348.9	13.2	115.2	881.3
3	641.6	192.3	9.8	28.8	870.1

kW, while other relevant parameters remain the same as in the IEEE 33-bus extended system case.

The training process of Method 1 using the DDPG-PPO algorithm is shown in Fig. 8. The changes in electricity prices are shown in Fig. 9. The training is conducted for 500 episodes, and the planning scheme converges. In a larger distribution network system, the bi-level game becomes more intense. The process still involves data collection and bi-level games until both agents converge. After more than 300 training episodes, the reward functions of both agents stabilize.

Tables V and VI show the planning scheme results obtained using different methods, along with the annual comprehensive benefits for each stakeholder. The results are similar to those

of the IEEE 33-bus extended system case, with Method 1 still showing a significant advantage. Under the bi-level game, Method 1 and Method 2 still maximize their own DG and ES grid-connected capacities to maximize their own interests. This stimulates DG and ES operators to actively participate in distribution network planning, which will increase the annual comprehensive revenue of DG and ES operators. The planning results help to increase the penetration rate of distributed power sources in the distribution network. Method 1 has a higher price response than Method 2, which improves the overall economic benefits.

VI. CONCLUSION

This paper proposes a SNLS coordination planning method based on RL and bi-level games for distribution network planning. Validated on IEEE 33-bus and 69-bus extended systems, the method effectively boosts the participation enthusiasm of all stakeholders, guarantees the economical and reliable operation of distribution networks, and greatly improves renewable energy accommodation capacity. By innovatively combining bi-level game and reinforcement learning, this work overcomes the limitations of conventional methods and optimizes the SNLS collaborative operation mechanism, providing an effective solution to existing technical challenges. It also supports adaptive planning for future distribution networks with high renewable penetration and flexible loads. From a global perspective, the proposed approach can be adapted to diverse grid structures and market rules, offering a valuable reference for the intelligent and low-carbon development of distribution networks worldwide.

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